

Explainable AI across Text and Vision

A reproducible comparison of LIME, SHAP, prototype retrieval and Grad-CAM, applied to every prediction of three classifiers

Predictions explained	5,561
Individual explanations generated	16,683
Total explanation compute	13.1 hours
Tasks	3 — Tweet Sentiment (3-class), Movie Review Sentiment (binary, counterfactually-augmented), Glasses Detection (binary)

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Abstract

This project applies four explanation methods — LIME, SHAP, prototype retrieval and Grad-CAM — to three classifiers spanning two modalities: a three-class tweet sentiment model, a binary movie-review model fine-tuned with LoRA, and a binary glasses detector built on ResNet-18. Accuracies are Tweet Sentiment 78.20%, Movie Review Sentiment 75.88%, Glasses Detection 99.96%.

The distinguishing property of this work is coverage. Explanations are not generated for a hand-picked handful of inputs; **every one of the 5,561 predictions in the evaluation sets carries a complete set of explanations**, yielding 16,683 individual explanations at a cost of 13.1 hours of compute. This makes it possible to report distributions rather than anecdotes: how often LIME's local surrogate actually fits, how often the retrieved prototypes contradict the prediction, how much the methods agree with one another, and what each costs.

Methods are ranked here on **faithfulness** — whether deleting what an explanation named actually moves the model — because that is the explainer's task: the model has already reached its conclusion, and the method's job is to account for how. Agreement with human annotation is reported alongside, but as a diagnostic rather than a verdict: it measures how human-like the *model* is, and a faithful explanation of a model that relies on something unhuman scores badly on it precisely because it is correct.

Four findings recur. First, the methods agree far more about the *sign* of a feature's contribution than about *which* features are worth naming, so any single top-k list overstates its own specificity. Second, on the text tasks explanation quality degrades where it is most needed: on the model's errors the local surrogates fit worst and the attributions shrink toward zero. Third, **perturbation-based explanation has a budget floor** — below roughly seven model evaluations per input feature, both LIME and SHAP fall to the performance of a random ranking, which is where the movie-review explanations sit; the gradient-based methods have no such floor and Grad-CAM records the highest faithfulness measured here from a single backward pass. Fourth, **plausible and faithful come apart**: on the image task the method whose strongest pixel most often lands inside the human-marked region is also the least faithful of the three, so a reviewer checking saliency maps by eye would be reassured by the method they should trust least.

Approach and reproducibility

The original version of this project was a set of Colab notebooks with uploaded checkpoints, hard-coded local paths and unseeded splits. It could not be re-run, and two of its analyses were computed against models that had not loaded correctly. This rebuild is a Python package with a single configuration file; every result in this report is regenerated by the commands below, in order.

```

pip install -r requirements.txt
python -m xai.train_glasses      # retrain the image model with a seeded split
python -m xai.run --all         # explain every prediction, resumably
python -m xai.verify --all      # fail loudly if anything is missing
python -m xai.summarize --all   # aggregate into summary.json
python -m xai.modelcard --task all # architecture, training, per-split accuracy
python -m xai.evaluate --task all # plausibility against the human annotation
python -m xai.faithfulness --task all # deletion, against a random-ranking control
python -m xai.behaviour --task all # what each method names; the budget analysis
python -m xai.report           # regenerate this document as HTML and PDF

```

The last three are optional: they are what produce the faithfulness, plausibility and behavioural chapters. Without them `xai.report` still builds, and simply omits those sections rather than failing.

Design decisions that affect the explanations

DECISION	WHY IT MATTERS
Attribution in pixel space	Image normalisation is applied inside the model's forward pass, not as preprocessing, so Grad-CAM, SHAP and LIME all attribute to pixels and are directly comparable. Attributing to normalised tensors would make the three maps describe different functions.
Gradient capture by tensor hook	Grad-CAM captures gradients with a hook on the activation tensor rather than <code>register_backward_hook</code> , which fires unreliably on modules with multiple inputs and silently produced wrong maps in the original code.
Tokenizer-aware masking for SHAP	Masked variants stay inside the model's own vocabulary rather than becoming arbitrary word deletions.
<code>bow=False</code> for LIME on text	Repeated words are treated as distinct positions, so negation scoping is not destroyed by the bag-of-words assumption.
Fixed SLIC segmentation	A fixed superpixel count keeps LIME's granularity comparable across images; the default segmenter varies with face size and made the original explanations incommensurable.
Every explanation records its own diagnostics	Surrogate R^2 , convergence delta and neighbour agreement are stored per prediction, so an explanation's trustworthiness is reported alongside its content rather than assumed.
Resumable, append-only runs	Each record is flushed to disk as it completes; a run of several hours can be interrupted and restarted without loss.

What is stored for each prediction

Nothing is truncated on disk. Every record holds the complete attribution, not a top-k slice of it, so any figure in this report can be recomputed from the artifacts and any claim in it can be checked against them.

METHOD	STORED FOR EVERY PREDICTION
LIME (text)	A signed weight for every word in the input, plus the number of perturbations used and the surrogate's R^2
SHAP (text)	A Shapley value for every token, plus the base value and the sum of the values, which is what makes the additivity property checkable
Prototypes	The nearest training examples with similarities, labels, and whether each agrees with the prediction
Grad-CAM	The full 224×224 map
SHAP (image)	The full 224×224 signed map and its convergence delta
LIME (image)	The full 224×224 map, the weight of every superpixel, and the segmentation used, so the explanation reconstructs exactly

Alongside these each record carries the model's full probability vector, the method's measured runtime and a path to its rendered card. `python -m xai.verify --all` re-reads everything that was written and fails if any sample is missing a record, any method errored, any explanation came back empty, or any referenced file is absent. The figures below are reported only for runs that pass it.

How the outputs are laid out

The output tree mirrors the input tree and preserves input order, and nothing is stored inside an archive. An image at `Dataset/Glasses/image/neg/00012.jpg` is explained at `explanations/neg/00012.html`, with its attribution maps in `saliency/neg/00012/` as one `.npz` per method rather than bundled into a `.npz` — which is a zip file, and cannot be inspected, diffed or partially read without unpacking it. Text cards carry their input row number, so listing the directory reproduces the order of the source CSV rather than a hash order.

Every prediction in either modality is readable as HTML. Image predictions previously produced only a rendered panel, which left the per-method numbers behind the picture — runtime, focus, peak location, convergence delta — visible only inside the JSONL.

Reading the explanations

Token weights are signed with respect to the *predicted* class throughout: a positive weight pushed the model toward the label it chose, a negative weight pushed away from it. In the rendered cards, teal marks support and amber marks opposition. For images, Grad-CAM maps are unsigned intensity; SHAP and LIME maps are signed and use a diverging scale centred on zero.

Tweet Sentiment (3-class)

Dataset

SPLIT	N	CLASS DISTRIBUTION	MEAN CHARS	MAX CHARS
train	24732	negative: 7003, neutral: 10005, positive: 7724	68	141
test	2748	negative: 778, neutral: 1112, positive: 858	68	141

Model and performance

BERT-base-uncased, fully fine-tuned, with a three-class head.

METRIC	VALUE
samples explained	2748
accuracy	0.7820
macro F1	0.7851
mean confidence when correct	0.9641
mean confidence when wrong	0.8847

CONFUSION MATRIX

TRUE \ PREDICTED	NEGATIVE	NEUTRAL	POSITIVE
negative	597	163	18
neutral	136	873	103
positive	23	156	679

CLASS	PRECISION	RECALL	F1	SUPPORT
negative	0.790	0.767	0.778	778
neutral	0.732	0.785	0.758	1112
positive	0.849	0.791	0.819	858

The model is better calibrated in the weak sense that it is on average 0.079 more confident when right than when wrong. This gap is small, so confidence alone is a poor filter for review.

How this model was trained

base model	bert-base-uncased
what was trained	full fine-tune — every parameter updated
optimizer	AdamW
learning rate	5e-05
epochs	3
batch size	16
max sequence length	128
validation split	0.1
seed	42
trained on	Google Colab GPU
per-epoch metrics	not captured
parameters	109,484,547 of 109,484,547 trained (100.00%)

ACCURACY ON EVERY SPLIT

SPLIT	N	ACCURACY
train	24,732	0.9715
test	2,748	0.7820

The gap between training and held-out accuracy is **18.9 percentage points** (97.2% → 78.2%). The model has memorised a substantial part of its training set, which is worth holding in mind when reading the explanations: an attribution can faithfully describe a decision that rests on memorisation rather than on a generalising feature.

Explanation methods

Every one of the 2748 predictions above carries an explanation from each of the 3 methods below. No sampling, no cherry-picking: the per-prediction records are in `artifacts/tweet/predictions.jsonl` and the rendered explanations in `artifacts/tweet/explanations/`.

METHOD	EXPLAINED	MEAN RUNTIME	TOTAL COMPUTE	FAILURES	SURROGATE R ²
lime	2748	4.59s	210 min	0.0%	0.573
shap	2748	1.29s	59 min	0.0%	—
prototype	2748	0.00s	0 min	0.0%	—

LIME — Local Interpretable Model-agnostic Explanations

LIME explains one prediction by perturbing the input many times, asking the model for a prediction on each perturbation, and fitting a weighted linear model to that local neighbourhood. The coefficients of that linear model are the explanation. It treats the classifier as a black box, so the same implementation works for a transformer and a CNN. Its central weakness is that the explanation is only as good as the local linear fit, which is why this project records the surrogate's R^2 alongside every LIME explanation rather than presenting the coefficients unqualified.

Across all 2748 explanations the local surrogate achieved a mean R^2 of 0.573, and fell below 0.3 on 23.5% of them. Those low- R^2 cases are exactly where a LIME bar chart would look authoritative while meaning very little. Mean cost 4.59s per prediction, 210 minutes to explain the whole set.

SHAP — SHapley Additive exPlanations

SHAP assigns each input feature its Shapley value: the average marginal contribution that feature makes to the prediction across all orderings in which features could be added. This is the unique attribution satisfying local accuracy, missingness and consistency. Exact computation is exponential, so approximations are used — a partition masker over tokens for text, and expected gradients between the input and a baseline for images. Both approximations report their own error, which this project records.

Mean cost 1.29s per prediction, 59 minutes to explain the whole set.

Prototype retrieval — example-based explanation

Rather than attributing to parts of the input, prototype explanation answers 'what does this remind the model of?'. Each input is embedded with the model's own [CLS] representation and the nearest training examples are retrieved by cosine similarity. The explanation is an analogy: this review was called positive because it sits next to these training reviews, which are positive. It offers no token-level detail, but it is the only one of the three that can expose the training data behind a decision.

On average 92.4% of retrieved neighbours carry the predicted label. For 0.9% of predictions *none* of the retrieved neighbours agreed with the model — decisions the training set does not visibly support. Mean cost 0.00s per prediction, 0 minutes to explain the whole set.

Does explanation quality hold up on the errors?

The reason to want post-hoc explanation is to understand predictions that cannot be taken on trust. It is therefore worth checking directly whether each method's own reliability diagnostic degrades on the predictions the model gets wrong. Each row splits one diagnostic by whether the prediction was correct.

DIAGNOSTIC	WHEN CORRECT	WHEN WRONG	DIFFERENCE
LIME surrogate R^2 — how well the local linear model fits	0.584 (n=2149)	0.532 (n=599)	+0.053
Prototype neighbour agreement	0.949 (n=2149)	0.835 (n=599)	+0.113
SHAP sum of token values — total attributed movement	0.588 (n=2149)	0.460 (n=599)	+0.128
Model confidence (baseline)	0.964 (n=2149)	0.885 (n=599)	+0.000

LIME's surrogate fit falls by 0.053 on the predictions the model gets wrong ($0.584 \rightarrow 0.532$). This is the uncomfortable result: the explanations are least well-grounded exactly on the predictions a human most needs explained. It is an argument for displaying the fit statistic alongside every explanation rather than presenting the coefficients alone.

Does the model use what the explanations named?

Comprehensiveness deletes the top-ranked features and measures how far the predicted-class probability falls; sufficiency keeps only those features. The random row ranks the same features arbitrarily at the same budget, and is the only thing that makes the other rows interpretable.

METHOD	COMPR. $\uparrow$	$\times$ RANDOM	SUFF. $\downarrow$	$\times$ RANDOM
lime	0.425	3.15 $\times$	0.146	0.35 $\times$
shap	0.428	3.17 $\times$	0.138	0.33 $\times$
random (baseline)	0.135	1.00 $\times$	0.418	1.00 $\times$

The strongest here is **shap** at 3.17 $\times$ a random ranking. That is comfortably above chance, so these attributions can be read as describing the model.

Does it point where a human points? SECONDARY

This compares each explanation against annotation produced by people rather than against the model. It describes how human-like *the model* is, seen through the explanation — so it is reported here as a diagnostic and does not decide which method explained better.

METHOD	PRECISION	RECALL	F1	IOU
lime	0.640	0.640	0.640	0.571
shap	0.569	0.569	0.569	0.504

Scored on 1,475 of 2,733 predictions; the 1,258 excluded are those whose human rationale is the entire input, where any method scores near 1.0 by construction.

What kind of feature does each method name?

Composition of each method's top-5 tokens across every prediction in this set. LIME perturbs whole words, so it can never return punctuation or a word fragment; SHAP works at wordpiece level and can. That asymmetry is structural, and it is what the lower rows measure.

CATEGORY	LIME	SHAP
content word	60.9%	43.5%
function word	28.3%	22.1%
sentiment word	9.8%	9.5%
sub-word fragment	0.3%	14.4%
punctuation	0.0%	9.8%
number	0.7%	0.7%

Counter-evidence

What each method says about a strongly polar word sitting in an input the model did *not* classify with that polarity. The two methods ask different questions: LIME asks whether deleting the word moves the prediction — for a word the model already discounted it does not, so the coefficient is near zero — while SHAP's additivity forces it to report the word as evidence the model overcame.

WORD	N	LIME	SHAP
good	35	0.0252	-0.0747
love	23	0.1048	-0.0509
bad	19	0.0764	-0.0241
sorry	15	0.1844	-0.0033
fun	14	0.0023	-0.1439
happy	13	0.0962	-0.0852
miss	13	-0.0241	-0.1716
sad	10	0.0927	-0.0254
sick	10	-0.0594	-0.1415
great	10	0.1117	0.0164

Contrast and negation

PROBE	LIME	SHAP
share of attribution after “but” (sentiment follows the second clause)	0.545	0.582
negator's rank, 0 = most important	0.376	0.389
negator's share of attribution mass	0.121	0.112

On the contrast probe the difference is +0.0368 (95% CI [+0.0132, +0.0597], $p = 0.0020$, $n = 255$), so SHAP does weight the clause after the marker more heavily. The negation rows show no such separation: **neither method surfaces the negator**, which is worth stating because it is the one place a bag-of-words assumption was expected to cost LIME something measurable.

Do the methods agree?

LIME and SHAP were compared on all 2748 predictions where both succeeded. Their top-5 token sets overlap with a mean Jaccard index of 0.392, they pick the same single most important token 50.1% of the time, and where they score a token in common they agree on its sign 65.1% of the time.

COMPARISON	VALUE
mean Jaccard of top-k tokens	0.392
same top-1 token	50.1%
sign agreement on shared tokens	65.1%
predictions compared	2748

Sign agreement is high while set overlap is moderate: the two methods rarely contradict each other about whether a word is evidence for or against the label, but they often disagree about which words are worth naming at all. Presenting either method's top-k list as 'the' explanation therefore overstates its specificity.

Worked examples

These are selected by measurable criteria, not by appearance: the most confident correct prediction, the most confident error, and the prediction whose own explanation is least well-grounded.

MOST CONFIDENT CORRECT PREDICTION

570eb26a35 • predicted **neutral** at 100.0% • correct

Hi fellow Peruvian

LIME (surrogate $R^2 = 0.80$)

Hi +0.00 Peruvian +0.00 fellow -0.00

SHAP

Hi +0.00 fellow +0.00 Peruvian +0.00

Nearest training examples

- sim 0.986 • **neutral** ✓: Hi Rob!
- sim 0.985 • **neutral** ✓: hřč½rlich!
- sim 0.983 • **neutral** ✓: Me hice una maruchan

MOST CONFIDENT ERROR — THE MODEL IS SURE AND WRONG7b69f49c42 · predicted **neutral** at 100.0% · **WRONG — true label positive**

tonight in party w/ my girls (minus vita)

LIME (surrogate $R^2 = 0.07$)

minus +0.00 in +0.00 party +0.00 tonight +0.00 my +0.00 girls -0.00 w -0.00
vita +0.00

SHAP

tonight -0.00 in -0.00 party -0.00 w -0.01 / +0.01 my -0.00 girls +0.00 (-0.00

Nearest training examples

- sim 0.988 · **neutral** ✓: My wife lost her sprint blackberry 8330 on a bike ride in AM. Anyone have an extra BB laying around? retweet
- sim 0.987 · **neutral** ✓: forecast in sf for friday, may 29: 66 degrees.
- sim 0.987 · **neutral** ✓: goodnight world.... goodnight eddplant.

WEAKEST LOCAL SURROGATE — THE EXPLANATION ITSELF IS UNRELIABLE99cebd8f8d8 · predicted **neutral** at 99.9% · **correct**

They grovelled On tomorrow morning, will post a link to the show, should be on just after 9am if anyone is about

LIME (surrogate $R^2 = 0.01$)

morning +0.00 post +0.00 if +0.00 On -0.00 9am +0.00 is -0.00 tomorrow +0.00
just +0.00

SHAP

They -0.00 grove -0.00 lled -0.00 On -0.00 tomorrow +0.00 morning -0.00 , +0.00
will +0.00

Nearest training examples

- sim 0.989 · **neutral** ✓: ohh yeahh with the showcase he said his artist will be outta town so keep me posted as to the next one
- sim 0.989 · **neutral** ✓: I was only @ Cambie just slightly before the event then had to leave Waiting for feedback though; I'll keep u posted!
- sim 0.986 · **neutral** ✓: Offline for now, I'll tweet again later...

CONFIDENT PREDICTION WITH NO SUPPORTING TRAINING NEIGHBOUR

5f12e0bb1a · predicted **neutral** at 93.5% · **WRONG — true label negative**

sorryyy milerz that i can` t vote i would love to vote but it doesn` t works i` m sad and very
frustrated :@ :@ --__--

LIME (surrogate $R^2 = 0.71$)

love +0.73 would +0.12 frustrated -0.12 i +0.06 sad -0.05 can +0.04 that +0.04
very -0.04

SHAP

sorry +0.01 y +0.01 y -0.02 mile -0.00 rz -0.00 that -0.00 i -0.00 can -0.00

Nearest training examples

- sim 0.984 · **negative** x: My #followillfriday sucks because while I was teaching people how to be nice, you **** were drooling over Notion
- sim 0.970 · **negative** x: lol. When I went to buy my new laptop in Feb., I already knew, lol, but I looked anyway and walked away sadfaced. ./
- sim 0.968 · **negative** x: it`s `s birthday today, wish her a good one assholes! Sleepytime

Findings

Explanations fail where the model fails

The pattern that recurs across the tweet set is that explanation quality tracks model confidence. On confident, correct predictions LIME's local surrogate fits well and both token methods converge on the same sentiment-bearing word. On the model's errors the surrogate R^2 collapses and SHAP's token values sum to approximately zero — the additive account says, correctly, that no token in the input pushed the model anywhere in particular.

This is worth stating plainly because it inverts the usual hope for post-hoc explanation. The explanations are least informative exactly on the predictions a human most needs explained. The practical consequence is that the surrogate fit statistic should be shown next to every explanation, not buried: it is the difference between “the model relied on this word” and “we could not establish what the model relied on”.

Movie Review Sentiment (binary, counterfactually-augmented)

Dataset

SPLIT	N	CLASS DISTRIBUTION	MEAN CHARS	MAX CHARS
train	1600	negative: 800, positive: 800	1954	2000
val	200	negative: 100, positive: 100	1940	2000
test	199	negative: 100, positive: 99	1964	2000

Model and performance

BERT-base-uncased with a LoRA adapter (rank 8, α 32, applied to the query and value projections of every attention layer) and a two-class head. Only the adapter and the head were trained.

Correction to the original project. The adapter checkpoint stores the fine-tuned classification head under keys prefixed `base_model.model.`. The original notebook loaded the adapter folder with `BertForSequenceClassification.from_pretrained`, which silently ignores those keys and leaves a randomly initialised head. Every LIME and SHAP explanation in the original movie-review analysis was therefore computed against a random classifier. This rebuild loads the adapter with `PeftModel.from_pretrained` and asserts that the head matches the checkpoint before any explanation is generated.

METRIC	VALUE
samples explained	199
accuracy	0.7588
macro F1	0.7581
mean confidence when correct	0.8036
mean confidence when wrong	0.6931

CONFUSION MATRIX

TRUE \ PREDICTED	NEGATIVE	POSITIVE
negative	70	30
positive	18	81

CLASS	PRECISION	RECALL	F1	SUPPORT
negative	0.795	0.700	0.745	100
positive	0.730	0.818	0.771	99

The model is better calibrated in the weak sense that it is on average 0.110 more confident when right than when wrong. This gap is large enough that confidence is a usable triage signal.

How this model was trained

base model	bert-base-uncased
what was trained	LoRA — adapter and classification head trained, base frozen
trained on	Google Colab GPU
optimizer	not recorded
learning rate	not recorded
epochs	not recorded
batch size	not recorded
per-epoch metrics	not captured
parameters	297,988 of 109,780,228 trained (0.27%)
LoRA adapter	rank 8, α 32, dropout 0.1, applied to the value and query projections of every attention layer
trainable breakdown	294,912 adapter + 3,076 classification head

ACCURACY ON EVERY SPLIT

SPLIT	N	ACCURACY
train	1,600	0.7662
val	200	0.7150
test	199	0.7588

Training and held-out accuracy differ by 0.7 percentage points (76.6% $\rightarrow$ 75.9%), so this model is not relying on memorisation to the degree its accuracy might otherwise suggest.

Explanation methods

Every one of the 199 predictions above carries an explanation from each of the 3 methods below. No sampling, no cherry-picking: the per-prediction records are in `artifacts/movie/predictions.jsonl` and the rendered explanations in `artifacts/movie/explanations/`.

METHOD	EXPLAINED	MEAN RUNTIME	TOTAL COMPUTE	FAILURES	SURROGATE R ²
lime	199	78.87s	262 min	0.0%	0.606
shap	199	39.21s	130 min	0.0%	—
prototype	199	0.00s	0 min	0.0%	—

LIME — Local Interpretable Model-agnostic Explanations

LIME explains one prediction by perturbing the input many times, asking the model for a prediction on each perturbation, and fitting a weighted linear model to that local neighbourhood. The coefficients of that linear model are the explanation. It treats the classifier as a black box, so the same implementation works for a transformer and a CNN. Its central weakness is that the explanation is only as good as the local linear fit, which is why this project records the surrogate's R^2 alongside every LIME explanation rather than presenting the coefficients unqualified.

Across all 199 explanations the local surrogate achieved a mean R^2 of 0.606, and fell below 0.3 on 7.5% of them. Those low- R^2 cases are exactly where a LIME bar chart would look authoritative while meaning very little. Mean cost 78.87s per prediction, 262 minutes to explain the whole set.

SHAP — SHapley Additive exPlanations

SHAP assigns each input feature its Shapley value: the average marginal contribution that feature makes to the prediction across all orderings in which features could be added. This is the unique attribution satisfying local accuracy, missingness and consistency. Exact computation is exponential, so approximations are used — a partition masker over tokens for text, and expected gradients between the input and a baseline for images. Both approximations report their own error, which this project records.

Mean cost 39.21s per prediction, 130 minutes to explain the whole set.

Prototype retrieval — example-based explanation

Rather than attributing to parts of the input, prototype explanation answers 'what does this remind the model of?'. Each input is embedded with the model's own [CLS] representation and the nearest training examples are retrieved by cosine similarity. The explanation is an analogy: this review was called positive because it sits next to these training reviews, which are positive. It offers no token-level detail, but it is the only one of the three that can expose the training data behind a decision.

On average 74.5% of retrieved neighbours carry the predicted label. For 7.0% of predictions *none* of the retrieved neighbours agreed with the model — decisions the training set does not visibly support. Mean cost 0.00s per prediction, 0 minutes to explain the whole set.

Does explanation quality hold up on the errors?

The reason to want post-hoc explanation is to understand predictions that cannot be taken on trust. It is therefore worth checking directly whether each method's own reliability diagnostic degrades on the predictions the model gets wrong. Each row splits one diagnostic by whether the prediction was correct.

DIAGNOSTIC	WHEN CORRECT	WHEN WRONG	DIFFERENCE
LIME surrogate R^2 — how well the local linear model fits	0.640 (n=151)	0.500 (n=48)	+0.141
Prototype neighbour agreement	0.837 (n=151)	0.458 (n=48)	+0.378
SHAP sum of token values — total attributed movement	0.302 (n=151)	0.219 (n=48)	+0.083
Model confidence (baseline)	0.804 (n=151)	0.693 (n=48)	+0.000

LIME's surrogate fit falls by 0.141 on the predictions the model gets wrong ($0.640 \rightarrow 0.500$). This is the uncomfortable result: the explanations are least well-grounded exactly on the predictions a human most needs explained. It is an argument for displaying the fit statistic alongside every explanation rather than presenting the coefficients alone.

Does the model use what the explanations named?

Comprehensiveness deletes the top-ranked features and measures how far the predicted-class probability falls; sufficiency keeps only those features. The random row ranks the same features arbitrarily at the same budget, and is the only thing that makes the other rows interpretable.

METHOD	COMPR. $\uparrow$	$\times$ RANDOM	SUFF. $\downarrow$	$\times$ RANDOM
lime	0.102	1.29 $\times$	0.197	0.85 $\times$
shap	0.088	1.12 $\times$	0.155	0.67 $\times$
random (baseline)	0.079	1.00 $\times$	0.230	1.00 $\times$

The strongest here is **lime** at 1.29 $\times$ a random ranking. That is barely above chance: deleting what the method ranked highest moves the model little more than deleting the same number of features at random, so these attributions should not be read as describing the model.

What kind of feature does each method name?

Composition of each method's top-5 tokens across every prediction in this set. LIME perturbs whole words, so it can never return punctuation or a word fragment; SHAP works at wordpiece level and can. That asymmetry is structural, and it is what the lower rows measure.

CATEGORY	LIME	SHAP
content word	72.4%	41.3%
function word	20.5%	24.1%
sentiment word	4.3%	0.7%
sub-word fragment	2.2%	9.1%
punctuation	0.0%	24.5%
number	0.6%	0.2%

Counter-evidence

What each method says about a strongly polar word sitting in an input the model did *not* classify with that polarity. The two methods ask different questions: LIME asks whether deleting the word moves the prediction — for a word the model already discounted it does not, so the coefficient is near zero — while SHAP's additivity forces it to report the word as evidence the model overcame.

WORD	N	LIME	SHAP
good	41	-0.0029	0.0014
bad	23	-0.0065	-0.0003
love	20	-0.0125	0.0016
best	18	-0.0021	0.0006
great	14	-0.0083	0.0012
lost	10	-0.0003	-0.0007
fun	9	0.0017	0.0012
nice	8	-0.0046	0.0011

Contrast and negation

PROBE	LIME	SHAP
share of attribution after “but” (sentiment follows the second clause)	0.661	0.679
negator's rank, 0 = most important	0.419	0.507
negator's share of attribution mass	0.010	0.008

On the contrast probe the difference is +0.0174 (95% CI [+0.0112, +0.0235], $p = 0.0001$, $n = 182$), so SHAP does weight the clause after the marker more heavily. The negation rows show no such separation: **neither method surfaces the negator**, which is worth stating because it is the one place a bag-of-words assumption was expected to cost LIME something measurable.

Do the methods agree?

LIME and SHAP were compared on all 199 predictions where both succeeded. Their top-5 token sets overlap with a mean Jaccard index of 0.012, they pick the same single most important token 0.0% of the time, and where they score a token in common they agree on its sign 53.6% of the time.

COMPARISON	VALUE
mean Jaccard of top-k tokens	0.012
same top-1 token	0.0%
sign agreement on shared tokens	53.6%
predictions compared	199

Sign agreement is high while set overlap is moderate: the two methods rarely contradict each other about whether a word is evidence for or against the label, but they often disagree about which words are worth naming at all. Presenting either method's top-k list as 'the' explanation therefore overstates its specificity.

Worked examples

These are selected by measurable criteria, not by appearance: the most confident correct prediction, the most confident error, and the prediction whose own explanation is least well-grounded.

MOST CONFIDENT CORRECT PREDICTION

movie_test_00131 · predicted **positive** at 94.8% · **correct**

like the great musical pieces of mozart himself , amadeus is a true work of art . it is one of those few movies of the 80 's that will be known for its class , its style , and its intelligence . why is this such a good film ? there are almost too many reasons to explain . the story : court composer salieri (f. murray abraham) feels waves of different emotions going through his head as wolfgang amadeus mozart (tom hulce) comes into his life as the young genius composer . salieri feels envy , and jealousy , but at the same time is fascinated with mozart 's brilliance and ingenious . we trave

LIME (surrogate $R^2 = 0.92$)

enables +0.01 annoying -0.01 f +0.01 this +0.01 emotion +0.01 amadeus +0.01
 young +0.01 brilliance +0.01

SHAP

like -0.00 the -0.00 great -0.00 musical -0.00 pieces -0.00 of -0.00 mozart -0.00
 himself -0.00

Nearest training examples

- sim 0.929 · **positive** ✓ : field of dreams almost defies description . although it 's about baseball , it 's not a baseball movie . although the film is extremely spiritual , it 's not about religion . and a
- sim 0.924 · **positive** ✓ : this christmas , little ralphie parker (peter billingsley) wants nothing more than a red ryder bb gun under the tree . although cautioned by many that he will ? shoot his eye out
- sim 0.916 · **positive** ✓ : near the end of frank capra 's holiday classic , its a wonderful life , george bailey (jimmy stewart) is feeling down . he has devoted his life to others , and never fulfilled hi

MOST CONFIDENT ERROR — THE MODEL IS SURE AND WRONG

movie_test_00132 · predicted **negative** at 90.9% · **WRONG — true label positive**

armageddon , in itself , symbolizes everything that is wrong in modern filmmaking . stories have been replaced with special effects ; character development gets overshadowed by bad dialogue ; plotting consists of a bunch of shit getting blown up . armageddon is as stupid , as loud and as shallow as any movie you 'll see come out this summer , or maybe even any other summer . but i loved every freaking minute of it . believe me , i 'm just as shocked as you are . hell , i do n't even know why i went to see it in the first place . the previews were so annoying that i predicted this was going to

LIME (surrogate $R^2 = 0.83$)

neither +0.03 mean +0.03 stupidest +0.03 911 +0.03 plotting +0.02 shallow +0.02
wrong +0.02 beam +0.02

SHAP

arm +0.00 aged +0.00 don +0.00 , +0.00 in +0.00 itself +0.00 , +0.00 symbol +0.00

Nearest training examples

- sim 0.932 · **negative** ✓: funny how your expectations can be defeated , and not in good ways . the ghost and the darkness promised (at least , it seemed to me to promise) a hemingwayesque showdown between
- sim 0.928 · **negative** ✓: the happy bastard 's quick movie review wild wild west a better name for this movie might 've been " wild wild waste " . warner bros . , in an attempt to get their own men in black
- sim 0.925 · **negative** ✓: cashing in on the success of white men ca n't jump and the failure of most of their solo vehicles (including the cowboy way and drop zone) , wesley snipes and woody harrelson hav

WEAKEST LOCAL SURROGATE — THE EXPLANATION ITSELF IS UNRELIABLE

movie_test_00027 · predicted **positive** at 56.6% · **WRONG — true label negative**

the lives of older people in the twilight of their years attempting to come to grips with their shared histories and possible futures is a fascinating topic . finding an all - star cast for such a film is a stroke of genius . combining all that with a three - time oscar - winning director (robert benton of " kramer vs. kramer ") and creating a decidedly mediocre movie is the stuff of disappointment . in yet another noir mystery set in hollywood -- how many of these have we seen during the past few years ? -- the atmosphere is moody , the actors enjoyable to watch and the story goes nowhere .

LIME (surrogate $R^2 = 0.17$)

detective -0.02 plot +0.02 of +0.02 become +0.02 i -0.02 one -0.02 that +0.02
crossroads +0.02

SHAP

the +0.00 lives +0.00 of +0.00 older +0.00 people +0.00 in +0.00 the +0.00
twilight -0.00

Nearest training examples

- sim 0.921 · **positive** ✓ : jake kasdan , son of one of the best screenwriters around , breaks into filmmaking by writing and directing this hard - to - categorize detective story : part mystery , part comedy
- sim 0.916 · **negative** ✗ : one would think that david duchovny , star of the cult favorite " x - files " would be very careful in choosing his leading man cinema roles . at least one would hope so . one would
- sim 0.904 · **positive** ✓ : the keen wisdom of an elderly bank robber , the naive ambitions of a sexy hospital nurse , and a partnership that blossoms between the two are the fine components that make up a mo

CONFIDENT PREDICTION WITH NO SUPPORTING TRAINING NEIGHBOUR

movie_test_00051 · predicted **positive** at 88.1% · **WRONG — true label negative**

the film may be called mercury rising , but that title does n't describe the trajectory taken by this motion picture , a routine thriller that combines government cover - ups with a cloying and poorly - motivated buddy story . the " hook " that is supposed to make mercury rising unique is that the young protagonist is autistic . however , aside from giving actor miko hughes a chance to win raves for his performance , this particular aspect of the film comes across as nothing more than a convenient plot device . those expecting to see even a semi - thorough exploration of the condition will be

LIME (surrogate $R^2 = 0.44$)

the +0.04 motivated +0.03 autistic -0.02 the +0.02 exploration +0.02 gold -0.02
that -0.02 but +0.02

SHAP

the -0.00 film -0.00 may -0.00 be +0.00 called +0.00 mercury +0.00
rising +0.00 , +0.00

Nearest training examples

- sim 0.912 · **negative** x : as a hot - shot defense attorney , kevin lomax (keanu reeves) has a special talent for picking juries that will ultimately acquit his clients . he is also an excellent judge of c
- sim 0.910 · **negative** x : 8 mm is not going to enlighten your day . it 's a dark , grimy and disturbing portrait of underworld snuff films and filled with excessive violence . not only do you need the stoma
- sim 0.910 · **negative** x : since director steven zaillian previously wrote the powerful screenplay for 1993 's steven spielberg drama " schindler 's list " and directed 1993 's intelligent " searching for bo

Findings**Where prototypes earn their place**

This is the weakest of the three models and the one where running more than one *kind* of explanation pays off. On reviews it gets wrong, the token methods tend to produce near-uniform attributions of very small magnitude — LIME's surrogate R^2 falls and SHAP's values shrink toward zero — so a reader looking only at a bar chart sees no warning. Prototype retrieval behaves differently: it returns training examples whose labels contradict the prediction outright, which is a visible, checkable signal that something is wrong.

A held-out example makes the point. For “The premise was promising but the pacing dragged and the dialogue felt wooden. I checked my watch twice.” the model answers **positive** at 77%. LIME's strongest token is “twice” at -0.043 , SHAP's is a full stop, and neither is interpretable. All three retrieved neighbours are negative reviews. The example-based method is the only one of the three that flags the error.

Glasses Detection (binary)

Dataset

2614 images, glasses: 1307, no_glasses: 1307, all resized to 224×224.

SPLIT	N	ACCURACY
test	392	1.0000
train	1830	1.0000
val	392	0.9974

Model and performance

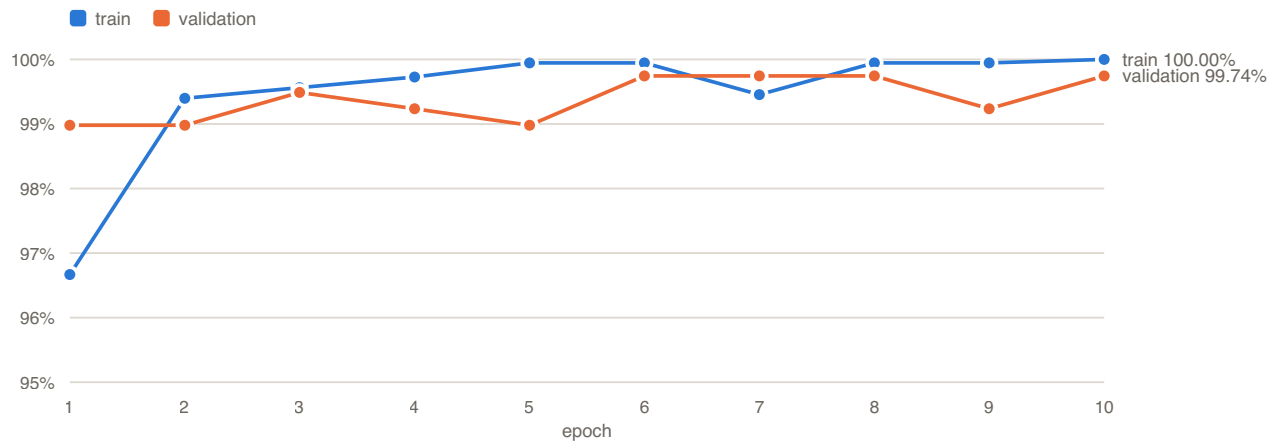
ResNet-18 initialised from ImageNet weights, final layer replaced with a 2-way linear head, fine-tuned for 10 epochs (Adam, lr 0.0001, batch 32) with horizontal flips, $\pm 10^\circ$ rotation and colour jitter. Split 1830/392/392, stratified and seeded (42). Training took 25.9 minutes on mps.

Correction to the original project. The original model was split with an unseeded `random_split`, so the held-out set cannot be reconstructed and its reported 93.88% validation accuracy cannot be verified. Measured across all 2614 images that checkpoint scores 99.96%, but roughly 80% of those images were in its own training set, so that figure is meaningless. The model was retrained here with a seeded stratified split written to `artifacts/glasses/split.json`. Its generalisation gap is +0.00 percentage points, against +5.88 for the original.

METRIC	VALUE
test accuracy	1.0000
best validation accuracy	0.9974
train accuracy (no augmentation)	1.0000
generalisation gap	+0.0000

Glasses detector — accuracy per epoch

y-axis starts at 95%, not zero, so the two series stay separable



METRIC	VALUE
samples explained	2614
accuracy	0.9996
macro F1	0.9996
mean confidence when correct	0.9989
mean confidence when wrong	0.9999

CONFUSION MATRIX

TRUE \ PREDICTED	NO_GLASSES	GLASSES
no_glasses	1306	1
glasses	0	1307

CLASS	PRECISION	RECALL	F1	SUPPORT
no_glasses	1.000	0.999	1.000	1307
glasses	0.999	1.000	1.000	1307

The model is barely calibrated in the weak sense that it is on average -0.001 more confident when right than when wrong. This gap is small, so confidence alone is a poor filter for review.

How this model was trained

train / val / test split	0.7 / 0.15 / 0.15
initialised from	True
epochs	10
batch size	32
learning rate	0.0001
weight decay	0.0001
augmentation	hflip 0.5, rotation 10, color_jitter 0.15
parameters	11,177,538 of 11,177,538 trained (100.00%)

ACCURACY ON EVERY SPLIT

SPLIT	N	ACCURACY
train	1,830	1.0000
val	392	0.9974
test	392	1.0000

Training and held-out accuracy differ by 0.0 percentage points (100.0% → 100.0%), so this model is not relying on memorisation to the degree its accuracy might otherwise suggest.

Explanation methods

Every one of the 2614 predictions above carries an explanation from each of the 3 methods below. No sampling, no cherry-picking: the per-prediction records are in `artifacts/glasses/predictions.jsonl` and the rendered explanations in `artifacts/glasses/explanations/`.

METHOD	EXPLAINED	MEAN RUNTIME	TOTAL COMPUTE	FAILURES	SURROGATE R ²
gradcam	2614	0.03s	1 min	0.0%	—
shap	2614	0.26s	11 min	0.0%	—
lime	2614	2.56s	112 min	0.0%	0.353

Grad-CAM — Gradient-weighted Class Activation Mapping

Grad-CAM takes the gradient of the predicted class score with respect to the final convolutional feature maps, averages those gradients spatially to get one weight per channel, and forms a weighted sum of the feature maps. The result is a coarse, class-discriminative heatmap showing which spatial regions drove the prediction. Its resolution is bounded by the feature map, here 7×7 upsampled to 224×224, so it localises regions rather than pixels.

The strongest 5% of pixels carry 16.6% of the total attribution mass on average, which is a direct measure of how focused this method's maps are. Mean cost 0.03s per prediction, 1 minutes to explain the whole set.

SHAP — SHapley Additive exPlanations

SHAP assigns each input feature its Shapley value: the average marginal contribution that feature makes to the prediction across all orderings in which features could be added. This is the unique attribution satisfying local accuracy, missingness and consistency. Exact computation is exponential, so approximations are used — a partition masker over tokens for text, and expected gradients between the input and a baseline for images. Both approximations report their own error, which this project records.

The mean convergence delta was 3.626. This is GradientSHAP's own estimate of how far its attributions are from summing to the difference in model output; larger values mean the additive account is approximate. The strongest 5% of pixels carry 35.1% of the total attribution mass on average, which is a direct measure of how focused this method's maps are. Mean cost 0.26s per prediction, 11 minutes to explain the whole set.

LIME — Local Interpretable Model-agnostic Explanations

LIME explains one prediction by perturbing the input many times, asking the model for a prediction on each perturbation, and fitting a weighted linear model to that local neighbourhood. The coefficients of that linear model are the explanation. It treats the classifier as a black box, so the same implementation works for a transformer and a CNN. Its central weakness is that the explanation is only as good as the local linear fit, which is why this project records the surrogate's R^2 alongside every LIME explanation rather than presenting the coefficients unqualified.

Across all 2614 explanations the local surrogate achieved a mean R^2 of 0.353, and fell below 0.3 on 28.0% of them. Those low- R^2 cases are exactly where a LIME bar chart would look authoritative while meaning very little. The strongest 5% of pixels carry 21.3% of the total attribution mass on average, which is a direct measure of how focused this method's maps are. Mean cost 2.56s per prediction, 112 minutes to explain the whole set.

Does explanation quality hold up on the errors?

The reason to want post-hoc explanation is to understand predictions that cannot be taken on trust. It is therefore worth checking directly whether each method's own reliability diagnostic degrades on the predictions the model gets wrong. Each row splits one diagnostic by whether the prediction was correct.

DIAGNOSTIC	WHEN CORRECT	WHEN WRONG	DIFFERENCE
LIME surrogate R^2 — how well the local linear model fits	0.353 (n=2613)	0.475 (n=1)	-0.123
Model confidence (baseline)	0.999 (n=2613)	1.000 (n=1)	+0.000

LIME's surrogate fit rises by 0.123 on the predictions the model gets wrong ($0.353 \rightarrow 0.475$). The fit does not degrade on errors, so a low-confidence prediction here is not automatically accompanied by a less trustworthy explanation.

Does the model use what the explanations named?

Comprehensiveness deletes the top-ranked features and measures how far the predicted-class probability falls; sufficiency keeps only those features. The random row ranks the same features arbitrarily at the same budget, and is the only thing that makes the other rows interpretable.

METHOD	COMPR. $\uparrow$	$\times$ RANDOM	SUFF. $\downarrow$	$\times$ RANDOM
gradcam	0.415	3.50 $\times$	0.159	0.35 $\times$
shap	0.300	2.53 $\times$	0.321	0.71 $\times$
lime	0.350	2.95 $\times$	0.244	0.54 $\times$
random (baseline)	0.119	1.00 $\times$	0.453	1.00 $\times$

The strongest here is **gradcam** at 3.50 $\times$ a random ranking. That is comfortably above chance, so these attributions can be read as describing the model.

Does it point where a human points? SECONDARY

This compares each explanation against annotation produced by people rather than against the model. It describes how human-like *the model* is, seen through the explanation — so it is reported here as a diagnostic and does not decide which method explained better.

METHOD	POINTING GAME	MASK AUC	IOU	MASS INSIDE MASK
gradcam	0.218	0.944	0.215	0.121
shap	0.704	0.785	0.126	0.129
lime	0.204	0.822	0.210	0.125

The human masks cover 4.4% of the frame, so a pixel chosen at random scores 0.044 on the pointing game, and a fixed centre pixel — no method at all — scores 0.095.

What shape is the evidence?

Connected components in the strongest 5% of each map, over 2,614 images. This measures what a focus score cannot: whether the evidence forms one region a reader can point at, or a field of scattered pixels.

METHOD	CONNECTED BLOBS	MASS IN LARGEST	VERTICAL CENTRE
gradcam	1.0	100.0%	119
shap	882.4	16.0%	108
lime	1.5	87.0%	111

Rows are 0 at the top of a 224-pixel frame; the eyes sit near row 95–110.

Do the methods agree?

Saliency maps were compared pairwise by Spearman rank correlation over the absolute attribution at every pixel — that is, do the methods agree about *where* the evidence is, setting polarity aside.

PAIR	IMAGES COMPARED	MEAN SPEARMAN ρ
gradcam_vs_shap	2614	0.488
gradcam_vs_lime	2614	0.316
shap_vs_lime	2614	0.209

The strongest agreement is gradcam vs shap ($\rho = 0.488$) and the weakest is shap vs lime ($\rho = 0.209$). The gap is structural rather than incidental: Grad-CAM and GradientSHAP are both gradient methods reading the same network, while LIME sees only input–output behaviour through superpixel occlusion and is additionally quantised to superpixel boundaries.

Worked examples

These are selected by measurable criteria, not by appearance: the most confident correct prediction, the most confident error, and the prediction whose own explanation is least well-grounded.

MOST CONFIDENT CORRECT PREDICTION

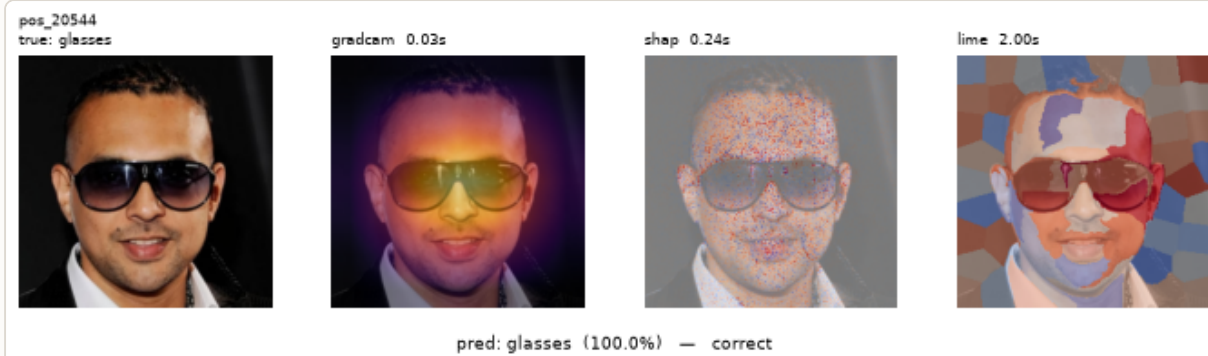
pos_00410 · predicted glasses at 100.0% · correct



METHOD	TOP-5% MASS	PEAK (Y,X)	RUNTIME
gradcam	0.140	[112, 111]	0.02s
shap	0.267	[102, 110]	0.24s
lime	0.209	[97, 107]	3.35s

MOST CONFIDENT ERROR — THE MODEL IS SURE AND WRONGneg_29801 · predicted glasses at 100.0% · **WRONG** — true label no_glasses

METHOD	TOP-5% MASS	PEAK (Y,X)	RUNTIME
gradcam	0.150	[112, 144]	0.02s
shap	0.330	[107, 140]	0.22s
lime	0.231	[87, 147]	1.89s

WEAKEST LOCAL SURROGATE — THE EXPLANATION ITSELF IS UNRELIABLEpos_20544 · predicted glasses at 100.0% · **correct**

METHOD	TOP-5% MASS	PEAK (Y,X)	RUNTIME
gradcam	0.147	[112, 112]	0.03s
shap	0.300	[102, 125]	0.24s
lime	0.122	[47, 153]	2.00s

Findings

What the single disagreement turned out to be

Across all 2,614 images the model disagrees with the ground-truth label exactly once, on `neg_29801`, which it calls *glasses* with 100.0% confidence. Opening the rendered panel settles the question immediately: **the subject is wearing glasses**. The image is filed under `neg/` and is mislabelled. Grad-CAM localises tightly on the frames and bridge, which is precisely the evidence a human would cite.

This is the most useful thing the image chapter produced, and it only appeared because every prediction was explained rather than a sample of them. The model's accuracy against the labels is 99.96%; against the images it is 100%. An explanation method here did not debug the model — it debugged the dataset. That is a routine use of XAI in practice and it is worth more than another heatmap of a correctly classified face.

Concentration is not coherence

The `top-5% mass` statistic reports SHAP as the more concentrated method, and taken alone it would rank the three the wrong way round for a human reader. The shape measurements above say why: SHAP is sparser *per pixel* while Grad-CAM is coherent *as a region*, and those are different properties that a single focus score collapses into one number. Counting connected components separates them, and the separation is not subtle — one contiguous blob against several hundred.

The consequence is the one this chapter is really about. SHAP's scattered peaks land inside the human-marked region more often than either rival's, so it is the method that looks most convincing when maps are checked by eye; it is simultaneously the least faithful of the three, so it is the method whose account of the model is least reliable. Those two facts are not in tension — they are what it means for plausibility and faithfulness to be different measurements — but a reader who knows only the first will draw exactly the wrong conclusion.

Which method explained best?

The model has already reached its conclusion. An explanation method's only task is to account for *how that model got there* — not to produce the account a human would have written. That distinction decides which of the measurements below counts as evidence, and it is worth stating before any number is quoted.

MEASUREMENT	COMPARES	WHAT IT ESTABLISHES
Faithfulness PRIMARY	explanation ↔ model	Whether the explainer did its job. Measured by deletion: remove what the method named and see whether the prediction moves.
Plausibility SECONDARY	explanation ↔ human	Whether the <i>model</i> is human-like. A property of the model, seen through the explanation — so it cannot rank explainers.
Self-reported fit	explanation ↔ itself	Internal consistency only: surrogate R^2 , convergence delta, neighbour agreement. Computed without any ground truth.

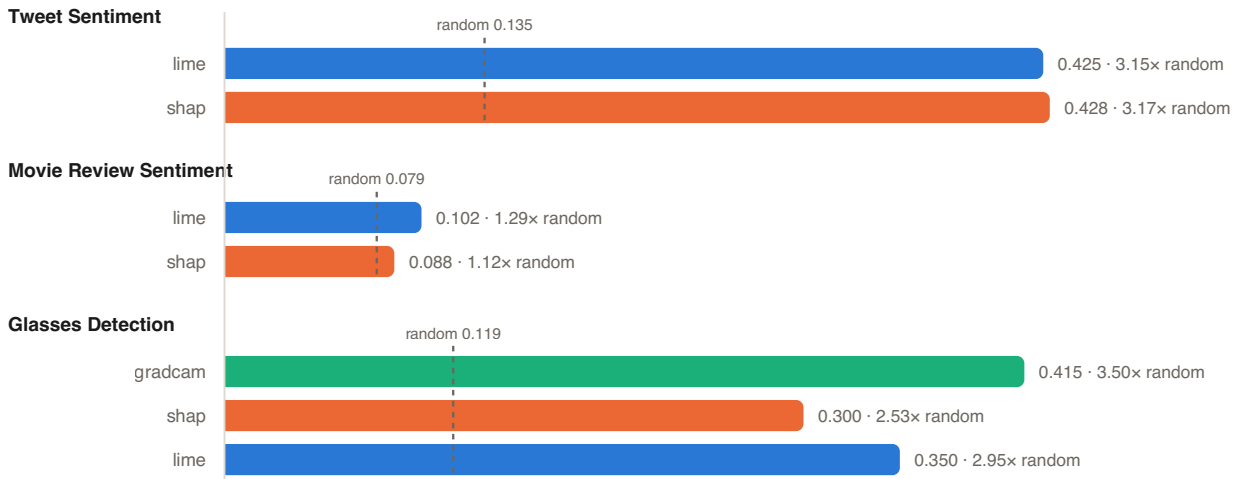
If a model latches onto something no human would — a punctuation habit, a spurious correlation — then a *faithful* explanation reports exactly that, and a human-agreement score penalises it for being correct. High plausibility with low faithfulness is therefore the worst combination in this study rather than the best: an explanation that reads convincingly while describing something the model never did.

Faithfulness — the primary criterion

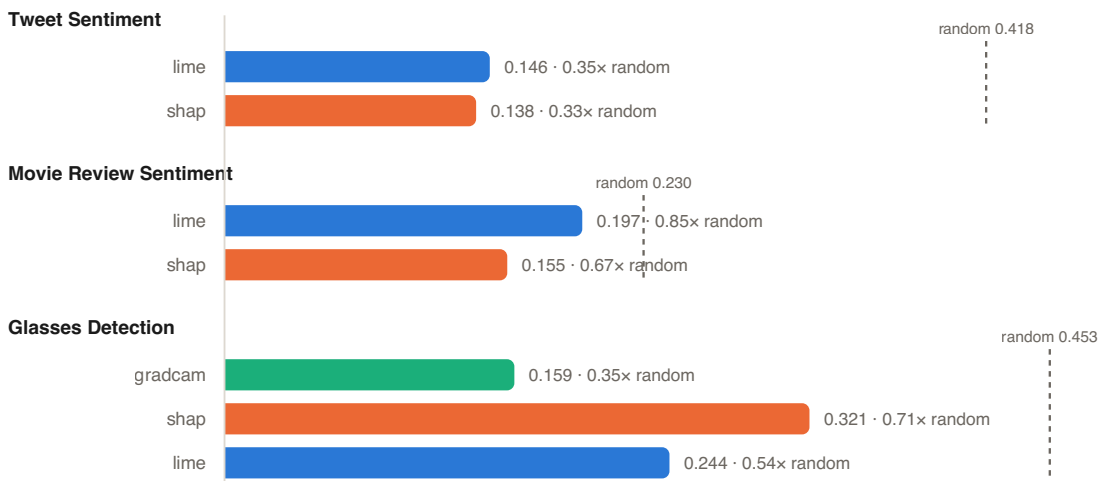
Comprehensiveness deletes the top-ranked features and asks how far the predicted-class probability falls; sufficiency keeps only those features and asks the same. Both are reported against a random ranking of the same features at the same budget, because deleting *any* fifth of an input moves a classifier — only the margin over random is evidence that the ordering carried information.

Faithfulness — comprehensiveness

delete the top-ranked features — how far the predicted-class probability falls · higher is better

**Faithfulness — sufficiency**

keep only the top-ranked features — how far it still falls · lower is better



FAITHFULNESS, ALL TASKS

TASK	METHOD	COMPR.	× RANDOM	SUFF.	× RANDOM	DELETIONS
Tweet Sentiment	lime	0.425	3.15×	0.146	0.35×	13,740
Tweet Sentiment	shap	0.428	3.17×	0.138	0.33×	13,740
Movie Review Sentiment	lime	0.102	1.29×	0.197	0.85×	995
Movie Review Sentiment	shap	0.088	1.12×	0.155	0.67×	995
Glasses Detection	gradcam	0.415	3.50×	0.159	0.35×	13,070
Glasses Detection	shap	0.300	2.53×	0.321	0.71×	13,070
Glasses Detection	lime	0.350	2.95×	0.244	0.54×	13,070

“deletions” counts the measurements behind each mean: one per prediction per budget, at $k = 1, 5, 10, 20$ and 50% of the input.

The verdicts

Each verdict below follows from the table above rather than from a reading of it: a lead smaller than 5% on comprehensiveness is recorded as a tie, and a task where the best method falls below $2\times$ a random ranking is recorded as having no usable winner.

TWEET SENTIMENT

Tie

shap leads lime by 0.8% on comprehensiveness, inside the 5% margin this report treats as indistinguishable. Both describe the model about equally well ($3.17\times$ and $3.15\times$ random).

MOVIE REVIEW SENTIMENT

Neither

every method here scores below $2\times$ a random ranking — the best, lime, reaches $1.29\times$. These explanations are close to noise, so there is nothing to choose between them.

GLASSES DETECTION

gradcam

gradcam leads lime by 18.4% on comprehensiveness and reaches $3.50\times$ a random ranking.

Plausibility — a secondary diagnostic

These figures compare each explanation against annotation produced by people: 2,614 attention masks shipped with the glasses dataset, and the human-selected rationale spans shipped with the tweet dataset. They are reported because they are informative about the models and because one result below matters a great deal — but they decide nothing, for the reason given at the head of this chapter. The movie-review dataset carries no human annotation, so plausibility is undefined there.

GLASSES — SALIENCY AGAINST HUMAN ATTENTION MASKS

METHOD	POINTING GAME	MASK AUC	IOU	MASS INSIDE MASK	N
gradcam	0.218	0.944	0.215	0.121	2,614
shap	0.704	0.785	0.126	0.129	2,614
lime	0.204	0.822	0.210	0.125	2,614

TWEET — TOKEN ATTRIBUTIONS AGAINST HUMAN RATIONALE SPANS

METHOD	PRECISION	RECALL	F1	IOU	N
lime	0.640	0.640	0.640	0.571	1,475
shap	0.569	0.569	0.569	0.504	1,475

Scored on 1,475 of 2,733 tweets. The 1,258 excluded are those whose `selected_text` is the entire tweet, where any method scores near 1.0 by construction; that exclusion is also why the scored sample is not class-balanced.

Plausible is not faithful

	FAITHFUL — THE MODEL USES IT	UNFAITHFUL — IT DOES NOT
MATCHES HUMAN ATTENTION	gradcam	shap
DOES NOT	lime	—

shap places its single strongest pixel inside the human-marked region 70.4% of the time — the best of the three, against the 4.4% a randomly chosen pixel achieves. It is also the *worst* of the three on both faithfulness metrics. Its peaks are where a human would look, but not what the model uses.

Ranked on human agreement it looks like the strongest method here; ranked on whether the model relies on what it names, it is the weakest. A reviewer checking saliency maps by eye would be reassured by precisely the method they should trust least — which is the practical reason plausibility is reported here as a diagnostic and never as a verdict.

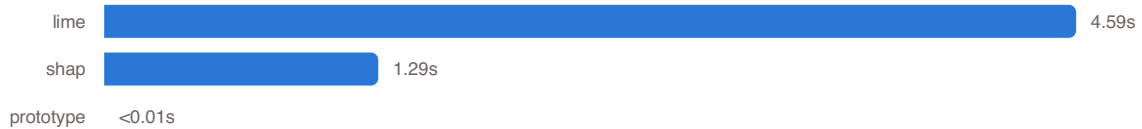
Cross-task comparison

Cost of explanation

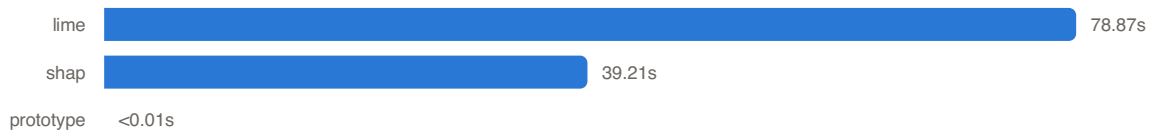
Cost of one explanation

mean wall-clock seconds per prediction · each task on its own scale

Tweet Sentiment



Movie Review Sentiment



Glasses Detection



TASK	METHOD	MEAN RUNTIME	N	FAILURE RATE
Tweet Sentiment	lime	4.59s	2748	0.0%
Tweet Sentiment	shap	1.29s	2748	0.0%
Tweet Sentiment	prototype	0.00s	2748	0.0%
Movie Review Sentiment	lime	78.87s	199	0.0%
Movie Review Sentiment	shap	39.21s	199	0.0%
Movie Review Sentiment	prototype	0.00s	199	0.0%
Glasses Detection	gradcam	0.03s	2614	0.0%
Glasses Detection	shap	0.26s	2614	0.0%
Glasses Detection	lime	2.56s	2614	0.0%

Averaged over tasks, prototype is the cheapest method at 0.00s per prediction and lime the most expensive at 28.67s, a factor of 17920. Cost matters for adoption: an explanation method that cannot keep up with the prediction rate can only ever be applied retrospectively to samples, which is precisely the practice this project set out to avoid.

The three classifiers

Each model was trained separately and the report has described them one at a time; set side by side, they differ on a dimension worth naming before the explanations are compared — how much of each network was actually fitted, and how far that shows up as memorisation.

TASK	ARCHITECTURE	PARAMETERS	TRAINED	TRAIN ACC.	TEST ACC.	GAP
Tweet Sentiment	BertForSequenceClassification	109,484,547	100.00%	0.9715	0.7820	+18.9 pp
Movie Review Sentiment	PeftModelForSequenceClassification	109,780,228	0.27%	0.7662	0.7588	+0.7 pp
Glasses Detection	ResNet	11,177,538	100.00%	1.0000	1.0000	+0.0 pp

Only one of the three carries a large generalisation gap: **Tweet Sentiment** at **+18.9 percentage points**, with 100% of its parameters updated over its training set. The others sit at Movie Review Sentiment +0.7 pp (0.27% of the network trained), Glasses Detection +0.0 pp (100.00% of the network trained). The contrast is not proof that adapter training generalises better — the tasks and dataset sizes differ too — but it does mean the explanations in each chapter are describing models of quite different character, and an attribution on the most-memorised model is the most likely to be faithfully reporting a decision that will not transfer.

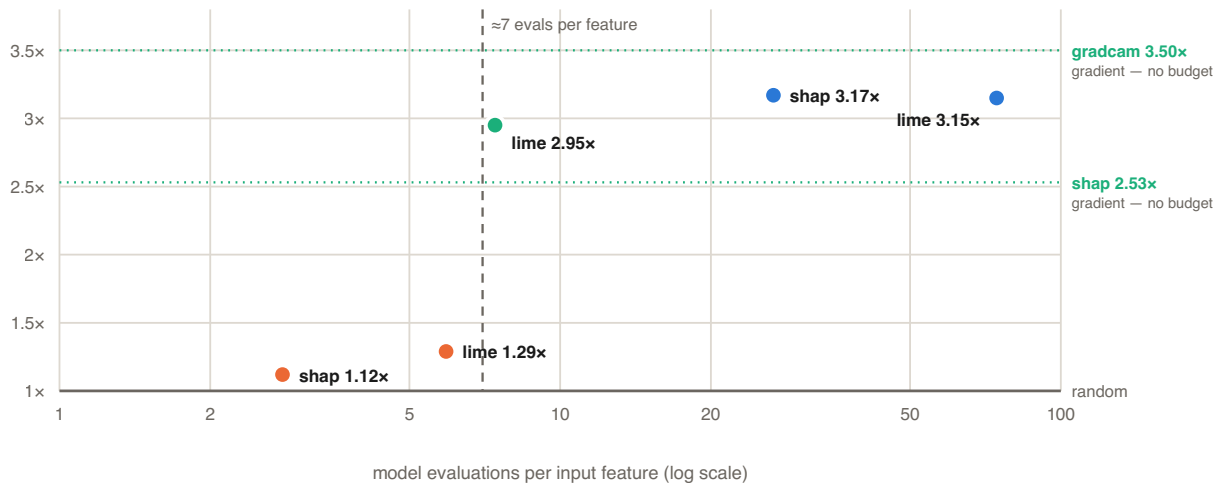
How much budget does an explanation need?

The cost above is spent buying model evaluations. What decides whether a perturbation method is *estimating* an attribution or interpolating one is not the total spend but how many of those evaluations each input feature receives — and that ratio varies by more than an order of magnitude across these three tasks.

How much budget does an explanation need?

faithfulness against evaluations per feature · perturbation methods only

■ tweet ■ movie ■ glasses



TASK	METHOD	FEATURES	BUDGET	EVALS / FEATURE	COMPR.	× RANDOM
Movie Review Sentiment	shap	430.1	1,200	2.8	0.088	1.12×
Movie Review Sentiment	lime	338.0	2,000	5.9	0.102	1.29×
Glasses Detection	lime	67.5	500	7.4	0.350	2.95×
Tweet Sentiment	shap	18.7	500	26.7	0.428	3.17×
Tweet Sentiment	lime	13.4	1,000	74.4	0.425	3.15×
Glasses Detection	gradcam	—	—	gradient	0.415	3.50×
Glasses Detection	shap	—	25	gradient	0.300	2.53×

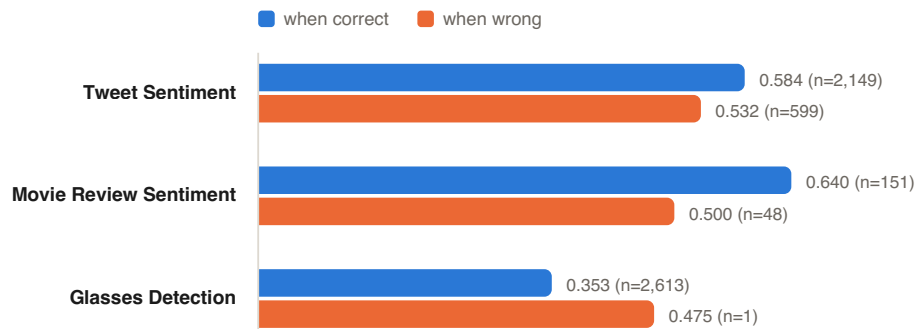
The perturbation methods separate cleanly on this axis. Every one that received at least 7.4 evaluations per feature clears 2× a random ranking; every one below 5.9 falls to near-random. The threshold drawn at ≈ 7 is declared rather than fitted — it is simply the gap between those two numbers, and 7 points would not support fitting one.

The gradient methods sit outside this relationship entirely. Grad-CAM reads the network directly in a single backward pass, takes no sampling budget at all, and records the highest faithfulness measured anywhere in this report. Set against the cost table above, that inverts the usual assumption: here the most expensive method on the most demanding task produced the least faithful explanations, and the cheapest produced the most faithful.

Explanation quality against model error

Does the explanation hold up on the model's errors?

mean LIME surrogate R^2 — how well the local linear model fits the classifier



On the text tasks the local surrogate fits less well where the model is wrong, which is the uncomfortable direction: the explanation is weakest exactly where a reader needs it most. It does not generalise, and the exception is worth naming rather than smoothing over — Glasses Detection (-0.123) runs the other way. That task has almost no errors to average over, so the comparison there rests on a handful of predictions and should not be read as a counter-example so much as an absence of evidence.

What each method is for

QUESTION BEING ASKED	METHOD	WHAT IT GIVES YOU	WHAT IT CANNOT TELL YOU	MEASURED FAITHFULNESS
Which words drove this decision?	SHAP	Signed per-token contributions with an additivity guarantee	Nothing about whether the training data supports the call	3.17× random at best (Tweet Sentiment), 1.12× at worst (Movie Review Sentiment)
Quick, model-agnostic sanity check	LIME	A local linear account of any black box, text or image	Reliability varies per sample; check the surrogate R^2 first	3.15× random at best (Tweet Sentiment), 1.29× at worst (Movie Review Sentiment)
What does this remind the model of?	Prototypes	The training examples standing behind the prediction, and their labels	No token-level detail at all	not defined — produces no per-feature attribution
Where in the image did it look?	Grad-CAM	A fast, class-discriminative region map	Pixel-level precision; it is bounded by a 7×7 feature grid	3.50× random on Glasses Detection

These are not substitutes. A token method and an example method answer different questions, and the cases where they disagree are the ones worth a human's attention.

DO THE METHODS FLAG THE SAME ERRORS?

TASK	PAIR	JACCARD	FLAGGED BY BOTH
Tweet Sentiment	lime $\cap$ shap	0.195	16
Tweet Sentiment	lime $\cap$ prototype	0.048	12
Tweet Sentiment	shap $\cap$ prototype	0.000	0
Movie Review Sentiment	lime $\cap$ shap	0.000	0
Movie Review Sentiment	lime $\cap$ prototype	0.116	5
Movie Review Sentiment	shap $\cap$ prototype	0.220	9

That claim is measurable, and it holds: on Tweet Sentiment the three signals together flag 46.6% of the model's errors against 34.9% for the best of them alone, and 197 errors are caught by prototype retrieval that neither token method flags; on Movie Review Sentiment the three signals together flag 89.6% of the model's errors against 85.4% for the best of them alone, and 27 errors are caught by prototype retrieval that neither token method flags. Pairing one attribution method with prototype retrieval therefore covers materially more than either alone — the overlap table shows they are not finding the same failures.

Limitations and honest caveats

1. **Post-hoc explanations are not the model's reasoning.** Every method here approximates a decision that was already made. A faithful-looking attribution is evidence about the function the network computes, not a transcript of how it computed it. This limitation is shared by all four methods and is not addressed by combining them.
2. **Agreement between methods is not correctness.** Where LIME and SHAP concur, they may be concurring on the same artefact. This report measures agreement because it is computable for every prediction, but agreement is a weaker property than faithfulness and should not be read as validation.
3. **Faithfulness is measured by deletion, and deletion is off-distribution.** Removing the top-ranked features produces an input the model has never seen: whitespace-stripped text, or an image with a region replaced by its own mean colour. A large probability drop therefore conflates “the model relied on this” with “the model has never seen an input shaped like this”. The random-ranking control absorbs much of that — it is equally off-distribution — but not all of it, and this is the main reason the comprehensiveness figures are compared with one another rather than read as absolute quantities.
4. **Plausibility covers two tasks, and one of them unevenly.** The movie-review dataset ships no human rationales, so plausibility is undefined there. On the tweet task the human span is the entire tweet for most neutral examples; those are excluded because any method scores near 1.0 on them, which leaves a scored sample that is not class-balanced. Both facts limit how far the plausibility numbers generalise — which is a further reason they are reported as a diagnostic rather than a verdict.
5. **The budget threshold is a hypothesis, not a law.** It rests on five perturbation measurements, one metric and three tasks, and is declared rather than fitted. What the data supports is that the two near-random cases received the fewest evaluations per feature and the three reliable ones the most; the specific value marks the gap between them and should not be transferred to another model or dataset without re-measuring.
6. **The behavioural probes use hand-built word lists.** The sentiment, function-word, negator and contrast-marker lists in `behaviour.py` were written by hand rather than taken from a lexicon resource. They are short enough to audit in full, and every membership decision is visible in that file, but they are neither exhaustive nor independently validated.
7. **The glasses task is close to saturated.** At the accuracy reported, there are very few errors to learn from, which limits how much the explanations can reveal about failure modes. A harder split — rimless frames, extreme poses, occlusion — would be more informative for XAI purposes than the current one.
8. **Single seed.** All results come from one seed. LIME in particular is stochastic, and its per-sample attributions vary between runs; the distributions reported here are stable but individual explanations should not be treated as exact.

Corrections to the original report

This rebuild contradicts the earlier version of this project on several points of fact, recorded here so the difference is not mistaken for a discrepancy in the data.

ORIGINAL CLAIM	WHAT THE DATA SHOWS
Glasses dataset contains 4,000 images, split 3,200 / 400 / 400	The dataset contains 2,614 images, 1,307 per class
Glasses model reaches 93.88% validation accuracy	Unverifiable: the split was unseeded and cannot be reconstructed. The model was retrained here with a seeded split to obtain a figure that can be checked
Movie-review LIME and SHAP explanations describe the BERT-LoRA model	They described a randomly initialised classification head; the adapter's trained head was never loaded
Explanations generated for "~60+ samples per method, per dataset"	This rebuild explains every prediction in every evaluation set
Method rankings presented as conclusions	The original rankings were assigned by inspection of a few dozen images. Every ranking here is computed: faithfulness by deletion against a random-ranking control, plausibility against the human annotation shipped with the datasets, and each verdict derived from those numbers rather than written next to them
SHAP's saliency described as the most focused, and read as the best	Concentration was measured, but coherence was not. Its strongest pixel does land inside the human-marked region most often of the three — and it is also the least faithful of the three, with its attribution mass scattered across hundreds of disconnected components rather than one region

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